COLIMITS, TENSOR PRODUCTS AND INTERNAL HOMS OF QUASI-COHERENT SHEAVES ON THE BIG SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author's own word 'the results shown above' occurs in the proof; it is quoted verbatim and refers to the earlier lemmas of the same section, which are delivered as stubs.

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Allowed standard prerequisites (editorial).

- The construction lands in quasi-coherent sheaves for each topology. Same chapter, Lemma lemma-properties-quasi-coherent-on-big, invoked at proof lines 3642-3644. Shows that the output of construction (1), (2) or (3) is quasi-coherent in each topology.
- The equivalence between quasi-coherent O_S-modules and quasi-coherent sheaves on the big site, given by the same formula for Zariski, etale and fppf. Same chapter, Proposition proposition-equivalence-quasi-coherent, invoked at proof lines 3646-3651. Supplies the common formula F -> F^a that makes the three topologies comparable.
- Compatibility of colimits and tensor products with the equivalence, and the topology-independence of finite local freeness. Same chapter, Lemmas lemma-equivalence-quasi-coherent-limits and lemma-equivalence-quasi-coherent-properties, invoked at proof lines 3658-3661, 3668-3672 and 3679-3691. Gives (1) and (2) and the parenthetical statement of (3).
- The construction of the internal hom of modules on a site depends only on the underlying presheaves. Modules on Sites, Section sites-modules-section-internal-hom, invoked at proof lines 3694-3700. Concludes (3): the internal homs computed in the three topologies agree.

Source reference. Original lemma 06VE: sources/stacks-descent/pinned-original-source.tex, source lines 3256–3256. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05VG: sources/stacks-descent/pinned-original-source.tex, source lines 2830–2830. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GNC: sources/stacks-descent/pinned-original-source.tex, source lines 3515–3515. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 03DX: sources/stacks-descent/pinned-original-source.tex, source lines 2716–2716. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 04TT: sources/stacks-chow/tags.txt, source lines 6266–6266. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme.*

1. *The category $QCoh((Sch/S)_{fppf}, \mathcal{O})$ has colimits and they agree with colimits in the categories $Mod((Sch/S)_{Zar}, \mathcal{O})$, $Mod((Sch/S)_{étale}, \mathcal{O})$, and $Mod((Sch/S)_{fppf}, \mathcal{O})$.*
2. *Given $\mathcal{F}, \mathcal{G}$ in $QCoh((Sch/S)_{fppf}, \mathcal{O})$ the tensor products $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G}$ computed in $Mod((Sch/S)_{Zar}, \mathcal{O})$, $Mod((Sch/S)_{étale}, \mathcal{O})$, or $Mod((Sch/S)_{fppf}, \mathcal{O})$ agree and the common value is an object of $QCoh((Sch/S)_{fppf}, \mathcal{O})$.*
3. *Given $\mathcal{F}, \mathcal{G}$ in $QCoh((Sch/S)_{fppf}, \mathcal{O})$ with $\mathcal{F}$ finite locally free (in fppf, or equivalently étale, or equivalently Zariski topology) the internal homs $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$ computed in $Mod((Sch/S)_{Zar}, \mathcal{O})$, $Mod((Sch/S)_{étale}, \mathcal{O})$, or $Mod((Sch/S)_{fppf}, \mathcal{O})$ agree and the common value is an object of $QCoh((Sch/S)_{fppf}, \mathcal{O})$.*

Proof. This lemma collects the results shown above in a slightly different manner. First of all, by Lemma 0GNC we already know the output of the construction in (1), (2), or (3) ends up in $QCoh((Sch/S)_\tau, \mathcal{O})$. It remains to show in each case that the result is independent of the topology used. The key to this is that the equivalence $QCoh(\mathcal{O}_S) \to QCoh((Sch/S)_\tau, \mathcal{O})$, $\mathcal{F} \mapsto \mathcal{F}^a$ of Proposition 03DX is given by the same formula independent of the choice of the topology $\tau \in \{Zariski, étale, fppf\}$.

Proof of (1). Let $\mathcal{I} \to QCoh((Sch/S)_{fppf}, \mathcal{O})$, $i \mapsto \mathcal{F}_i$ be a diagram. Write $\mathcal{F}_i = \mathcal{G}_i^a$ so we get a diagram $\mathcal{I} \to QCoh(\mathcal{O}_S)$. Then $\operatorname{colim} \mathcal{F}_i = (\operatorname{colim} \mathcal{G}_i)^a$ in $Mod((Sch/S)_\tau, \mathcal{O})$ for $\tau \in \{Zariski, étale, fppf\}$ by Lemma 06VE. This proves (1).

Proof of (2). Write $\mathcal{F} = \mathcal{H}^a$ and $\mathcal{G} = (\mathcal{I})^a$ with $\mathcal{H}$ and $\mathcal{I}$ quasi-coherent on S. Then $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G} = (\mathcal{H} \otimes_{\mathcal{O}} \mathcal{I})^a$ in $Mod((Sch/S)_\tau, \mathcal{O})$ for $\tau \in \{Zariski, étale, fppf\}$ by Lemma 06VE. This proves (2).

Proof of (3). Let $\mathcal{F}$ and $\mathcal{G}$ be in $QCoh((Sch/S)_{fppf}, \mathcal{O})$. Write $\mathcal{F} = \mathcal{H}^a$ with $\mathcal{H}$ quasi-coherent on S. By Lemma 05VG we have

$$\begin{aligned} \mathcal{F} \text{ finite locally free in fppf topology} &\Leftrightarrow \mathcal{H} \text{ finite locally free on } S \\ &\Leftrightarrow \mathcal{F} \text{ finite locally free in étale topology} \\ &\Leftrightarrow \mathcal{H} \text{ finite locally free on } S \\ &\Leftrightarrow \mathcal{F} \text{ finite locally free in Zariski topology} \end{aligned}$$

This explains the parenthetical statement of part (3). Now, if these equivalent conditions hold, then $\mathcal{H}$ is finite locally free. The construction of $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$ in Modules on Sites, Section 04TT depends only on $\mathcal{F}$ and $\mathcal{G}$ as presheaves of modules (only whether the output $\mathcal{H}om$ is a sheaf depends on whether $\mathcal{F}$ and $\mathcal{G}$ are sheaves). □